Classification

Logistic Regression, LDA, QDA, Naive Bayes & KNN — ISLR Chapter 4

Machine Learning with R

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Week 2, 2025

Today's Roadmap

1. Why not linear regression?
2. Logistic Regression
3. LDA & QDA
4. Naive Bayes
5. KNN for classification
6. Confusion matrix & ROC
7. Comparing methods

Datasets:

- `Default` — credit card default
- `Smarket` — S&P 500 direction
- `Weekly` — weekly returns
- `Boston` — crime rate

Methods:
Logistic, LDA, QDA, NB, KNN

The Classification Problem

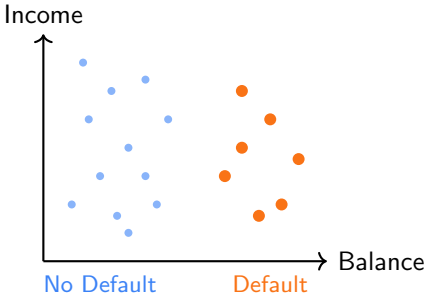
Given:
$(x_1, y_1), \ldots, (x_n, y_n)$
where $y_i \in \{1, 2, \ldots, K\}$

Goal:
Build a classifier $C(x)$ that assigns a new observation x to one of K classes

Often: First estimate

$$P(Y = k \mid X = x)$$

then classify to the most probable class

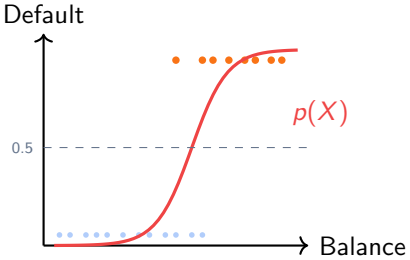


A Real Example: Credit Card Default

Default dataset:
$n = 10{,}000$ customers

Variable	Type
`default`	Yes/No
`student`	Yes/No
`balance`	\$0–\$2,654
`income`	\$772–\$73,554

Default rate $\approx 3.3\%$
Imbalanced classes!



Why Not Linear Regression?

Problem 1: Arbitrary coding

If $Y \in \{\text{stroke}, \text{OD}, \text{epilepsy}\}$

Code as $Y = 1, 2, 3$?

Linear regression implies:

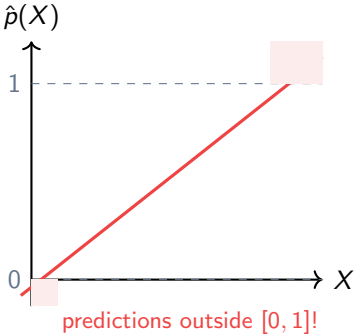
OD is "between" stroke and epilepsy

Ordering is meaningless!

Problem 2: Out-of-range

$\hat{Y}$ can be < 0 or > 1

Not a valid probability!



Solution: Logistic regression

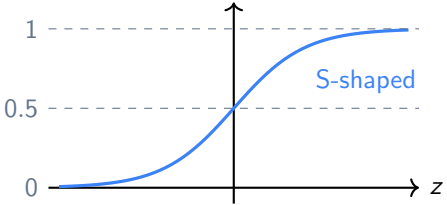
Outputs are always in $(0, 1)$

Logistic Regression: The Model

Key idea: Model $P(Y = 1 \mid X)$ directly, ensure it stays in $(0, 1)$

$$p(X) = P(Y = 1 \mid X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

The logistic (sigmoid) function:



Properties:

- $p(X) \in (0, 1)$ always
- $p(X) = 0.5$ when $\beta_0 + \beta_1 X = 0$
- S-shaped curve
- Monotonic: as $X \uparrow$, p moves toward 0 or 1

The Odds and the Log-Odds

Odds:

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$$

$p(X)$	Odds
0.2	1:4
0.5	1:1
0.8	4:1
0.9	9:1

Log-odds (logit):

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X$$

The **logit** is linear in X!

β_1: a one-unit increase in X changes the **log-odds** by β_1

Not the probability itself —
the relationship is non-linear

Estimating Coefficients: Maximum Likelihood

No closed-form solution (unlike OLS)

Likelihood function:

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \cdot \prod_{i:y_i=0} (1 - p(x_i))$$

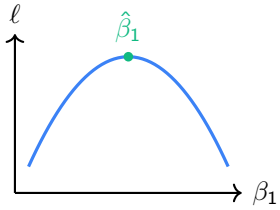
Choose $\hat{\beta}_0, \hat{\beta}_1$ to **maximize** ℓ

Solved numerically (Newton–Raphson)

R does it for you: `glm(..., family = binomial)`

Intuition:

Find coefficients that make the **observed data most probable**



Worked Example: Default Data

```
glm(default ~ balance, family =
binomial)
```

	$\hat{\beta}$	SE	p
Intercept	-10.65	0.36	< 0.001
balance	0.0055	0.0002	< 0.001

Interpretation:
Each \$1 increase in balance $\Rightarrow$
log-odds of default increases by 0.0055

Predict at balance = \$1000:

$$\hat{p} = \frac{e^{-10.65+0.0055\times 1000}}{1+e^{-10.65+0.0055\times 1000}}$$

$$= \frac{e^{-5.15}}{1+e^{-5.15}} = 0.0058$$

0.6% chance of default

At balance = \$2000:

$\hat{p} = 0.586$ (58.6%!)

Multiple Logistic Regression

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

Default: all predictors

	$\hat{\beta}$	p-value
Intercept	-10.87	< 0.001
balance	0.0057	< 0.001
income	0.0030	0.62
student	-0.6468	< 0.001

Income is not significant!

Surprise: Student is negative!

In simple logistic: student is *positive*

In multiple logistic: student is *negative*

Why?

Students have higher *balances*

Controlling for balance, students are *less* likely to default

Confounding — same as Newspaper!

Making Predictions: The Decision Boundary

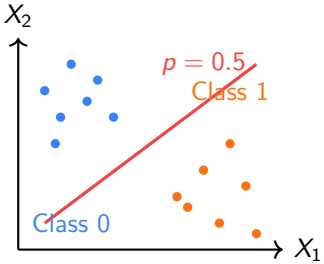
Default rule:
Classify to "Yes" if $\hat{p}(X) \geq 0.5$

Equivalently: classify to "Yes" if

$$\hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots \geq 0$$

The **decision boundary** is the set of X where $\hat{p}(X) = 0.5$

For logistic regression: it's a **linear** boundary



Changing the Threshold

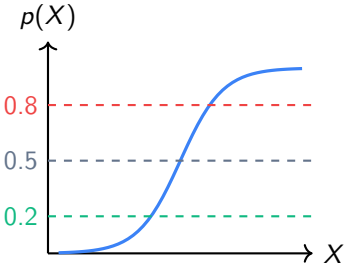
What if we use a threshold other than 0.5?

Lower threshold (e.g., 0.2):

- More "Default" predictions
- Catches more true defaults
- But more false alarms

Higher threshold (e.g., 0.8):

- Fewer "Default" predictions
- Misses many true defaults
- But very few false alarms



The right threshold depends on the **cost** *of each type of error*

The Confusion Matrix

		Predicted	
		Negative	Positive
Actual	Negative	TN	FP
	Positive	FN	TP

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$ (true positive rate)

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$ (true negative rate)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{n}$$

Accuracy can be misleading with imbalanced classes!

Worked Example: Default Confusion Matrix

Threshold $= 0.5$, test set $n = 10{,}000$

	Pred. No	Pred. Yes
Actual No	9,627	40
Actual Yes	228	105

Accuracy $= (9627 + 105)/10000 = 97.3\%$

Sounds great?

But look closer:

Sensitivity $= 105/(105 + 228) = 31.5\%$

We miss 68.5% of defaulters!

Why?

Only 3.3% default — predicting "No" always gives 96.7% accuracy!

$\Rightarrow$ Lower the threshold to catch more defaults

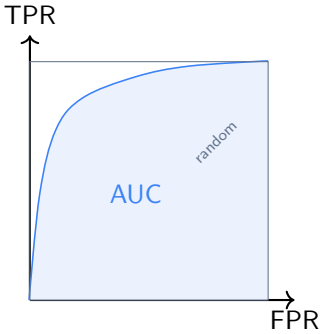
The ROC Curve

ROC = Receiver Operating Characteristic

Plot **Sensitivity** vs $\mathbf{1} - \mathbf{Specificity}$
for all possible thresholds

AUC (Area Under Curve):

- AUC $= 0.5$: random guessing
- AUC $= 1.0$: perfect classifier
- AUC > 0.8: good
- AUC > 0.9: excellent



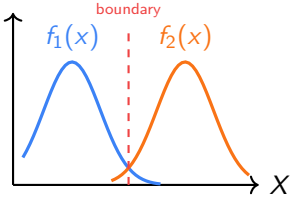
Generative Models: A Different Approach

Logistic regression: models $P(Y \mid X)$ directly
Generative approach: models $P(X \mid Y)$ for each class, then uses Bayes' theorem

Bayes' Theorem:

$$P(Y = k \mid X = x) = \frac{\pi_k \cdot f_k(x)}{\sum_{l=1}^{K} \pi_l \cdot f_l(x)}$$

$\pi_k = P(Y = k)$ — prior probability
$f_k(x) = P(X = x \mid Y = k)$ — class density



If we know each class's distribution, we can compute $P(Y \mid X)$ exactly!

Linear Discriminant Analysis: Idea

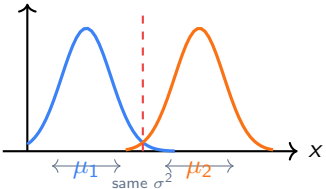
Assumption:
$X \mid Y = k \sim N(\mu_k, \sigma^2)$
Each class is Gaussian with
same variance σ^2, different means μ_k

Discriminant function:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

Classify to the class with largest $\delta_k(x)$

Linear in $x \Rightarrow$ **Linear** Discriminant



LDA: Worked Example ($p = 1$, $K = 2$)

Data:

Class 1: $n_1 = 30$, $\hat{\mu}_1 = 2$

Class 2: $n_2 = 20$, $\hat{\mu}_2 = 5$

$\hat{\sigma}^2 = 1.5$, $n = 50$

$\hat{\pi}_1 = 30/50 = 0.6$

$\hat{\pi}_2 = 20/50 = 0.4$

Discriminant functions:

$$\hat{\delta}_1(x) = x \cdot \frac{2}{1.5} - \frac{4}{3} + \log(0.6)$$
$$= 1.33x - 1.33 - 0.51 = 1.33x - 1.84$$

$$\hat{\delta}_2(x) = x \cdot \frac{5}{1.5} - \frac{25}{3} + \log(0.4)$$
$$= 3.33x - 8.33 - 0.92 = 3.33x - 9.25$$

Boundary: Set $\hat{\delta}_1(x) = \hat{\delta}_2(x)$:

$$1.33x - 1.84 = 3.33x - 9.25 \quad \Rightarrow \quad x = 3.71$$

Classify to Class 1 if $x < 3.71$, Class 2 if $x > 3.71$

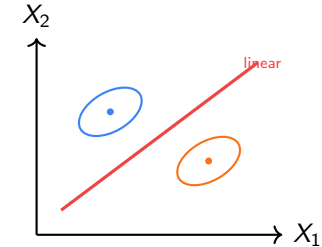
LDA with $p > 1$: Multivariate

Assumption: $X \mid Y = k \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$

$$\delta_k(x) = x^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k + \log(\pi_k)$$

Key points:

- **Common** covariance $\boldsymbol{\Sigma}$ for all classes
- Decision boundary is **linear** (hyperplane)
- Estimates: $\hat{\boldsymbol{\mu}}_k$, $\hat{\boldsymbol{\Sigma}}$, $\hat{\pi}_k$
- R: `MASS::lda(y ~ x1 + x2)`



same shape = linear boundary

Quadratic Discriminant Analysis (QDA)

Relaxation: Each class has its **own** covariance matrix $\mathbf{\Sigma}_k$

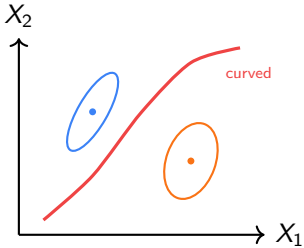
$$\delta_k(x) = -\frac{1}{2}(x-\mu_k)^T \mathbf{\Sigma}_k^{-1}(x-\mu_k) - \frac{1}{2}\log|\mathbf{\Sigma}_k| + \log \pi_k$$

Decision boundary is **quadratic** (curved)

Bias-Variance:

- LDA: Kp parameters (lower variance)
- QDA: $Kp(p+1)/2$ parameters (more flexible)

QDA better when classes truly have different covariance structures



different shapes = curved boundary

LDA vs. QDA: When to Use Which?

Feature	LDA	QDA
Covariance	Common $\mathbf{\Sigma}$	Class-specific $\mathbf{\Sigma}_k$
Boundary	Linear	Quadratic
# Parameters	Kp	$Kp(p+1)/2$
Variance	Lower	Higher
Bias	Higher	Lower
Best when	Small n, similar classes	Large n, different shapes

Same bias-variance trade-off as always!

R: `MASS::lda(...)` vs `MASS::qda(...)`

Naive Bayes Classifier

Key assumption: Predictors are **independent** within each class

$$f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$$

Why "naive"?
Independence is almost never true!
But it works surprisingly well in practice

Advantage:

- Estimate p one-dimensional densities instead of one p-dimensional density
- Works well with large p
- Very fast

Options for f_{kj}:

X_j type	f_{kj}
Quantitative	Gaussian
Quantitative	Kernel density
Qualitative	Frequency table

R: `e1071::naiveBayes(...)`

KNN for Classification (Revisited)

$$P(Y = k \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} \mathbf{1}(y_i = k)$$

No model assumptions at all!

- Completely non-parametric
- K controls flexibility
- Decision boundary can be very complex
- Struggles with large p (curse of dimensionality)

Smarket data: KNN results

K	Test accuracy
1	50.0%
3	53.2%
5	51.6%
10	52.0%

Not great! Stock data is very noisy

R: `class::knn(...)`

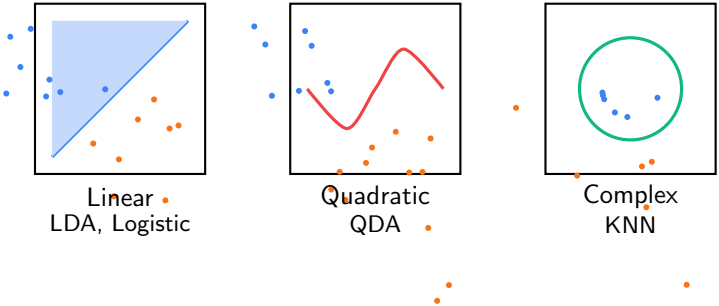
Comparing All Five Methods

	Logistic	LDA	QDA	NB	KNN
Boundary	linear	linear	quadratic	depends	flexible
Assumptions	logit linear	Gaussian, Σ	Gaussian, Σ_k	independence	none
# Params	$p+1$	Kp	$Kp^2/2$	Kp	0
Interpretable	✓✓	✓	$\sim$	$\sim$	×
Large p	good	good	risky	great	bad

Rule of thumb:

- Linear boundary true? $\Rightarrow$ Logistic $\approx$ LDA
- Moderate non-linearity? $\Rightarrow$ QDA or Naive Bayes
- Small p, large n, complex boundary? $\Rightarrow$ KNN

When Does Each Method Win?



Smarket Data: Method Comparison

S&P 500 daily returns: Predict Up/Down using Lag1, Lag2
Train: 2001–2004 Test: 2005

Method	Test Accuracy	Note
Logistic Regression	56.1%	best linear method
LDA	56.0%	almost identical
QDA	59.9%	best overall
Naive Bayes	59.0%	close to QDA
KNN ($K = 1$)	50.0%	random coin flip

QDA wins $\Rightarrow$ evidence of non-linear decision boundary
Logistic $\approx$ LDA $\Rightarrow$ both produce linear boundaries
KNN overfits badly with $K = 1$

Logistic Regression: Interpreting Coefficients

$\hat{\beta}_1 = 0.0055$ (balance)

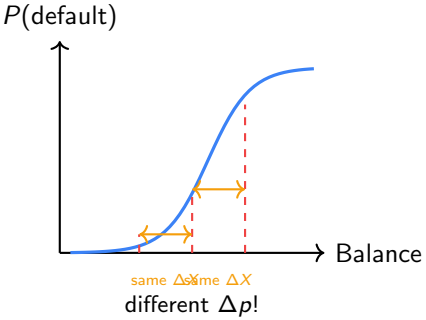
Odds ratio:

$$e^{\hat{\beta}_1} = e^{0.0055} = 1.0055$$

A \$1 increase in balance multiplies the odds of default by 1.0055

A \$100 increase:
$e^{0.0055 \times 100} = e^{0.55} = 1.73$

Odds increase by 73%!



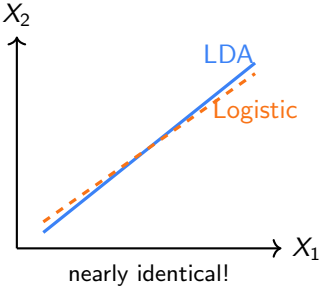
Why LDA and Logistic Give Similar Results

Both produce linear decision boundaries

LDA: assumes Gaussian classes with $\mathbf{\Sigma}$ shared
Logistic: directly models log-odds as linear

When they differ:

- LDA better if Gaussian assumption holds
- Logistic more robust to non-Gaussian data
- LDA uses full density $\Rightarrow$ more efficient if correct
- Logistic uses only boundary $\Rightarrow$ more robust



Multiclass Logistic: Softmax

For $K > 2$ classes:

$$P(Y = k \mid X) = \frac{e^{\beta_{k0} + \beta_{k1} X_1 + \cdots + \beta_{kp} X_p}}{\sum_{l=1}^{K} e^{\beta_{l0} + \beta_{l1} X_1 + \cdots + \beta_{lp} X_p}}$$

- Each class k has its own coefficient vector $\boldsymbol{\beta}_k$
- Probabilities sum to 1 across classes
- Called **softmax** function
- Used heavily in deep learning (Week 9)

Example: $K = 3$ classes
$p_1 = 0.6,\ p_2 = 0.3,\ p_3 = 0.1$
$\sum = 1.0$ ✓
Classify to Class 1

Poisson Regression: Count Data

$$\log(\lambda(X)) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

where $Y \sim \text{Poisson}(\lambda)$

When to use:

- Response is a **count** (0, 1, 2, ...)
- Examples: # bike rentals, # ER visits
- Cannot be negative

Bikeshare data:

```
bikers ~ hr + temp + weathersit
```

Generalized Linear Models:

Response	Link
Continuous	identity (OLS)
Binary	logit (logistic)
Count	log (Poisson)

R: `glm(..., family = poisson)`
All are special cases of GLMs!

Prior Probabilities Matter

LDA uses π_k in the discriminant function:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

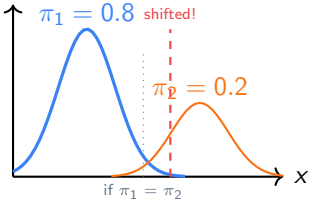
If $\pi_1 \gg \pi_2$: boundary shifts toward Class 2

Imbalanced classes:

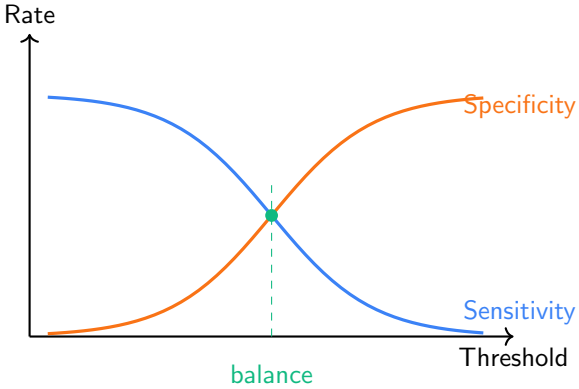
Default rate $= 3.3\%$

$\pi_{\text{No}} = 0.967$, $\pi_{\text{Yes}} = 0.033$

LDA is biased toward "No"

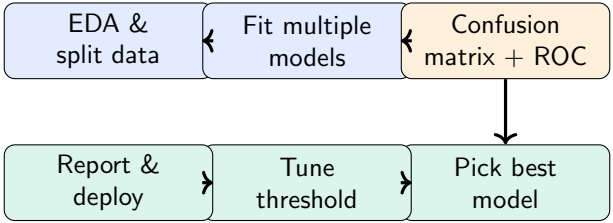


Sensitivity vs. Specificity: The Trade-Off



As threshold ↑: sensitivity ↓, specificity ↑
The ROC curve traces this entire trade-off

Practical Workflow for Classification



Always compare: Logistic, LDA, QDA, KNN on the same test set!

R Functions: Quick Reference

`glm(y ~ x, family = binomial)`	Logistic regression
`predict(fit, type = "response")`	Predicted probabilities
`MASS::lda(y ~ x, data)`	Linear discriminant analysis
`MASS::qda(y ~ x, data)`	Quadratic discriminant analysis
`e1071::naiveBayes(y ~ x)`	Naive Bayes
`class::knn(train, test, cl, k)`	K-nearest neighbors
`table(predicted, actual)`	Confusion matrix

`contrasts(factor_var)`	See dummy variable coding
`predict(lda.fit)$posterior`	Posterior probabilities
`predict(lda.fit)$class`	Predicted classes

Key Formulas: Cheat Sheet

Logistic Regression:

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$$

$$\text{logit}(p) = \log\frac{p}{1-p} = \beta_0 + \beta_1 X$$

LDA ($p = 1$):

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log\pi_k$$

Bayes' Theorem:

$$P(Y = k \mid X) = \frac{\pi_k f_k(x)}{\sum_l \pi_l f_l(x)}$$

Evaluation:

$$\text{Sens.} = \text{TP}/(\text{TP}+\text{FN})$$
$$\text{Spec.} = \text{TN}/(\text{TN}+\text{FP})$$
$$\text{Prec.} = \text{TP}/(\text{TP}+\text{FP})$$

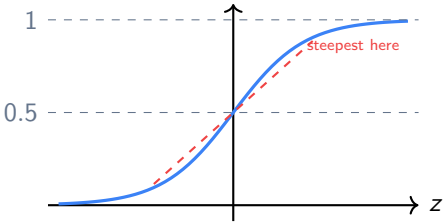
Think About It: Why Does the Sigmoid Work?

We need: $p(X) \in (0, 1)$

Start with linear predictor:
$z = \beta_0 + \beta_1 X \in (-\infty, +\infty)$

Apply sigmoid: $\sigma(z) = \dfrac{e^z}{1 + e^z}$

z	$\sigma(z)$
$-\infty$	0
-2	0.12
0	0.50
$+2$	0.88
$+\infty$	1



Smoothly maps $\mathbb{R} \to (0, 1)$

Worked Example: Will This Customer Default?

Model: $\log\left(\frac{p}{1-p}\right) = -10.65 + 0.0055 \cdot \text{balance}$

Customer A: balance $= \$500$
$z = -10.65 + 0.0055 \times 500 = -7.90$
$p = \sigma(-7.90) = 0.0004$
0.04% chance — safe!

Customer B: balance $= \$1{,}500$
$z = -10.65 + 0.0055 \times 1500 = -2.40$
$p = \sigma(-2.40) = 0.083$
8.3% chance — watch closely

Customer C: balance $= \$2{,}000$
$z = -10.65 + 0.0055 \times 2000 = 0.35$
$p = \sigma(0.35) = 0.587$
58.7% chance — likely default!

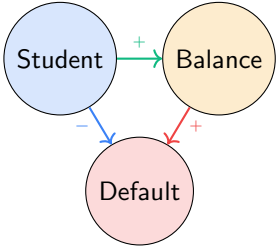
Customer D: balance $= \$2{,}500$
$z = -10.65 + 0.0055 \times 2500 = 3.10$
$p = \sigma(3.10) = 0.957$
95.7% — almost certain default

The Confounding Story: Student Status

Simple logistic:
default $\sim$ student
$\hat{\beta}_{\text{student}} = +0.40$ ($p < 0.001$)
Students default *more*!

Multiple logistic:
default $\sim$ balance + student
$\hat{\beta}_{\text{student}} = -0.65$ ($p < 0.001$)
Students default *less*!

The explanation:



Students carry higher balances but at same balance, they're more responsible $\Rightarrow$ lower default

Maximum Likelihood: Why Not Least Squares?

Least squares:

- Minimizes $\sum(y_i - p(x_i))^2$
- Works, but is not optimal
- Can give multiple solutions
- Non-convex optimization

Maximum likelihood:

- Maximizes $\prod p_i^{y_i}(1-p_i)^{1-y_i}$
- Unique global maximum
- Convex (in log-likelihood)
- Statistically efficient

Log-likelihood:

$$\ell = \sum_{i=1}^{n} [y_i \log p_i + (1-y_i)\log(1-p_i)]$$

This is also called **cross-entropy loss**

Used everywhere in deep learning!

MLE and cross-entropy are the same thing with different names

LDA: The Geometry

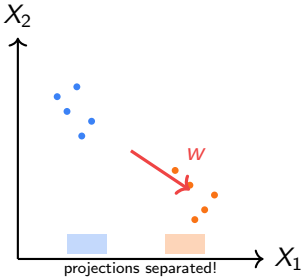
What LDA actually does:

1. Project data onto the direction that **maximizes** separation between class means
2. While **minimizing** within-class variance
3. This direction is $\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$

Fisher's criterion:

$$J = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}$$

Maximize between-class / within-class

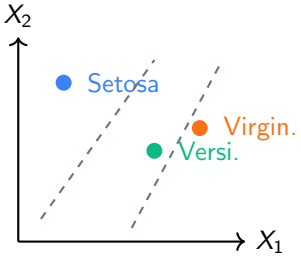


LDA for $K = 3$ Classes: Worked Example

Three types of iris flowers: Setosa, Versicolor, Virginica

	n_k	$\hat{\mu}_{k1}$	$\hat{\mu}_{k2}$
Setosa	50	5.01	3.43
Versicolor	50	5.94	2.77
Virginica	50	6.59	2.97

Using sepal length (X_1) and sepal width (X_2)
$\hat{\pi}_k = 1/3$ for all classes



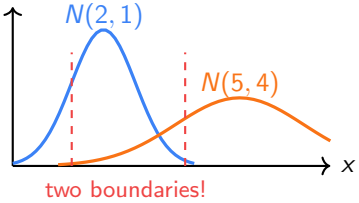
Two linear boundaries for 3 classes

QDA: Worked Numerical Example

Two classes, $p = 1$:

Class 1: $\hat{\mu}_1 = 2$, $\hat{\sigma}_1^2 = 1$
Class 2: $\hat{\mu}_2 = 5$, $\hat{\sigma}_2^2 = 4$
$\hat{\pi}_1 = \hat{\pi}_2 = 0.5$

Decision boundary:
Where $\delta_1(x) = \delta_2(x)$
This gives a **quadratic** equation in x



Different variances $\Rightarrow$ two crossing points

Naive Bayes: Worked Example

Spam filter: Is this email spam?

Features: contains "free", "click", "dear"
$\hat{\pi}_{\text{spam}} = 0.4$, $\hat{\pi}_{\text{ham}} = 0.6$

	free	click	dear
$P(\cdot \mid \text{spam})$	0.8	0.7	0.1
$P(\cdot \mid \text{ham})$	0.1	0.1	0.5

Email has: free=Yes, click=Yes, dear=No

Computation:

$P(\text{spam}) \times P(\text{free}|\text{spam}) \times P(\text{click}|\text{spam}) \times P(\overline{\text{dear}}|\text{spam})$
$= 0.4 \times 0.8 \times 0.7 \times 0.9 = 0.202$

$P(\text{ham}) \times P(\text{free}|\text{ham}) \times P(\text{click}|\text{ham}) \times P(\overline{\text{dear}}|\text{ham})$
$= 0.6 \times 0.1 \times 0.1 \times 0.5 = 0.003$

$P(\text{spam}|X) = \frac{0.202}{0.202+0.003} = 98.5\%$

Spam!

Confusion Matrix: Why Accuracy Is Misleading

Disease screening:
1% of patients have the disease

"Always predict healthy":
Accuracy = 99%!
But sensitivity = 0% — misses everyone

Lesson:
With imbalanced classes, use:

- Sensitivity + Specificity
- Precision + Recall
- F1 score
- AUC

F1 Score:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Harmonic mean of precision & recall

Metric	Value
Accuracy	99%
Sensitivity	0%
F1	0%

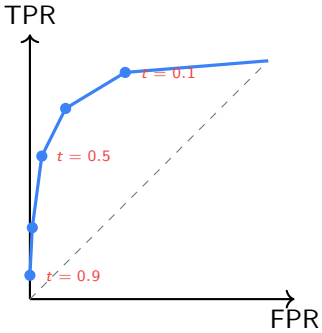
F1 exposes the problem!

ROC Curve: Step-by-Step Construction

For each threshold t:

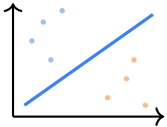
1. Classify: predict "Yes" if $\hat{p} \geq t$
2. Compute TPR (sensitivity)
3. Compute FPR (1− specificity)
4. Plot the point (FPR, TPR)

Threshold	TPR	FPR
0.1	0.95	0.40
0.3	0.80	0.15
0.5	0.60	0.05
0.7	0.30	0.01
0.9	0.10	0.00



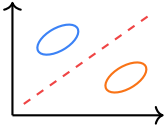
LDA vs. Logistic: Same Boundary, Different Path

Logistic (discriminative):
Models $P(Y \mid X)$ directly
learns the boundary



Only cares about the **boundary**
More robust when Gaussian assumption fails

LDA (generative):
Models $P(X \mid Y)$ for each class
learns the distributions



Models the full **distribution** of each class
More efficient when assumption holds

In practice, they almost always give the same answer!

Number of Parameters: A Practical Guide

Method	# Parameters	Example: $p = 50$, $K = 3$
Logistic	$(K-1)(p+1)$	$2 \times 51 = 102$
LDA	$Kp + p(p+1)/2$	$150 + 1275 = 1425$
QDA	$Kp + Kp(p+1)/2$	$150 + 3825 = 3975$
Naive Bayes	$2Kp$	300
KNN	0	0

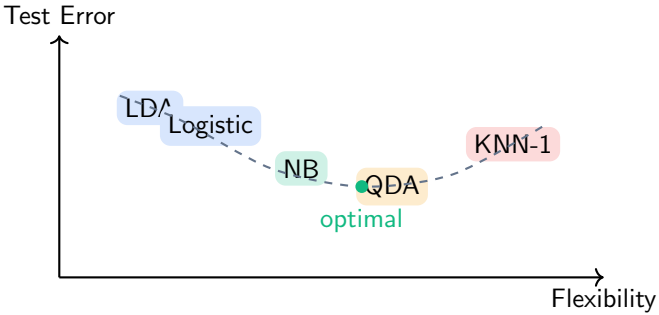
Key insight:

QDA estimates **separate** covariance matrices $\Rightarrow$ many parameters

With $p = 50$: QDA needs ~ 4000 parameters!

$\Rightarrow$ QDA is risky when n is not much larger than p^2

The Bias-Variance Trade-Off for Classification



Same principle: more flexible $\neq$ always better!

Threshold Selection: A Business Decision

Credit card fraud:

	Cost of FP	Cost of FN
Fraud	Block card ($10)	Lose $5,000
Default	Deny credit ($200)	Lose $10,000
Cancer	Biopsy ($500)	Miss cancer

When FN is expensive:

Lower the threshold (e.g., 0.1)

Catch more true positives

Accept more false positives

Expected cost:

$$C = C_{\text{FP}} \times \text{FP} + C_{\text{FN}} \times \text{FN}$$

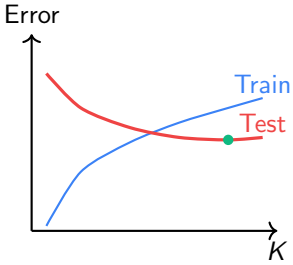
Choose threshold that minimizes C

The optimal threshold depends on the relative cost of errors, not just accuracy!

KNN: Choosing K on the Default Data

K	Train Error	Test Error
1	0%	4.2%
3	2.1%	3.5%
5	2.5%	3.1%
10	2.8%	2.9%
50	3.0%	2.8%
100	3.2%	2.7%

Best test error at $K \approx 100$



Note: $1/K$ is the flexibility
(small K = more flexible)

Logistic Regression: Assumptions

What logistic regression assumes:

1. Log-odds is **linear** in X
2. Observations are **independent**
3. No **perfect multicollinearity**
4. Sample size is **adequate**
   (rule of thumb: ≥ 10 events per predictor)

What it does NOT assume:

- Normal distribution of X
- Constant variance
- Linear relationship between X and Y

Checking assumptions:

- Linearity of logit: plot log-odds vs. X
- VIF for collinearity
- Deviance residuals
- Hosmer-Lemeshow test

Logistic is more flexible than LDA — fewer distributional assumptions

Deviance: The Logistic Version of RSS

In linear regression: quality = RSS
In logistic regression: quality = **Deviance**

$$D = -2\sum_{i=1}^{n}\left[y_i \log(\hat{p}_i) + (1-y_i)\log(1-\hat{p}_i)\right]$$

- Lower deviance = better fit
- Analogous to RSS
- Can compare nested models: $\Delta D \sim \chi^2_{\Delta p}$
- Null deviance: model with intercept only
- Residual deviance: our model

R output:

```
Null deviance:  2920.6
Residual deviance:  1596.5
AIC: 1600.5
```

Deviance dropped by 1324
$\Rightarrow$ balance is very informative!

Multinomial Logistic vs. LDA for $K > 2$

Multinomial logistic:

- Directly models $P(Y = k \mid X)$
- No distributional assumptions
- Softmax function
- More robust

R: `nnet::multinom(y ~ x)`

LDA ($K > 2$):

- Assumes Gaussian within each class
- Estimates K means + shared $\mathbf{\Sigma}$
- More efficient if assumptions hold
- Provides nice visualizations

R: `MASS::lda(y ~ x)`

For $K = 2$: Use `glm(family=binomial)` — simpler and standard

The Role of Priors: A Bayesian Perspective

$$P(Y = k \mid X) = \frac{\pi_k \cdot f_k(x)}{\sum_l \pi_l \cdot f_l(x)}$$

π_k = **prior** probability

Default: use sample proportions

$\hat{\pi}_k = n_k/n$

But you can adjust!

- Know disease prevalence? Set π_k accordingly
- Want to be conservative? Increase prior for "dangerous" class

Example:

Default rate = 3.3%

With $\hat{\pi}_{\text{Yes}} = 0.033$:
Boundary at balance $\approx$ \$1,800

With $\pi_{\text{Yes}} = 0.5$ (equal priors):
Boundary at balance $\approx$ \$1,200

More conservative: catches more defaulters at cost of more false positives

Cross-Validation Preview: How to Choose the Best Method

Problem:
We compared 5 methods on Smarket but used a **single** train/test split

Results depend on which split!

Solution (Week 3):
***k*-fold cross-validation**

1. Split data into k folds
2. Train on $k-1$, test on 1
3. Repeat k times
4. Average test errors

5-fold CV

More reliable estimate of test error!

Quick Quiz: Test Yourself!

1. What is $P(\text{default})$ if the log-odds is -3?

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   $p = e^{-3}/(1 + e^{-3}) = 0.0474 \approx 4.7\%$

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   Sensitivity decreases, Specificity increases

4. A classifier always predicts "No" on imbalanced data. Accuracy $= 97\%$. Good?

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   No — sensitivity = 0%, it misses all positives. Check AUC or F1.

Quick Quiz: More Questions

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   Exactly 0% — each point is its own nearest neighbor

8. ROC curve hugs the top-left corner. Is this good or bad?

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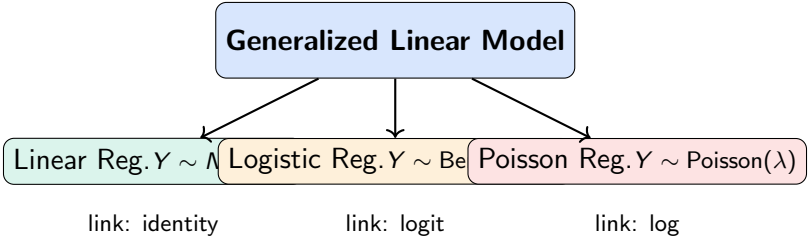
   Good — high TPR with low FPR at many thresholds. AUC ≈ 1

Common Mistakes in Classification

1. **Using accuracy on imbalanced data**
   - Always report sensitivity, specificity, AUC
2. **Evaluating on training data**
   - Always use a held-out test set or CV
3. **Ignoring the threshold**
   - 0.5 is default, not optimal. Tune it!
4. **Coding $K > 2$ classes as numbers for linear regression**
   - Use logistic / LDA / softmax instead
5. **Forgetting to standardize for KNN**
   - KNN uses distances — scale matters!

GLM Family: The Bigger Picture



All three: $g(E[Y]) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$

R: `glm(y ~ x, family = gaussian / binomial / poisson)`

Summary: Chapter 4 Key Takeaways

- Don't use linear regression for classification
- Logistic regression: model $P(Y \mid X)$ via sigmoid
- Coefficients: change in **log-odds** per unit X
- MLE estimation (not least squares)

- LDA: Gaussian classes, shared $\boldsymbol{\Sigma}$
- QDA: Gaussian classes, separate $\boldsymbol{\Sigma}_k$
- Naive Bayes: independence assumption
- KNN: non-parametric, curse of dimensionality
- ROC + AUC for evaluation

Next week: How do we properly estimate test error? $\Rightarrow$ Cross-Validation!

Exercise 1: Default Data (ISLR 4.8 Q13)

Data: `Default` $n = 10{,}000$

1. Fit logistic regression: `default ~ income + balance`
2. Split into train/test (use `set.seed(1)`)
3. Compute test error rate
4. Repeat with different random splits — does error vary?
5. Add `student` as a predictor — does it improve?

Hint: Test error should be around 2.5–3.5%

Exercise 2: Smarket Data (ISLR 4.8 Q10)

Data: `Smarket` Predict: `Direction`

1. EDA: Produce numerical and graphical summaries
2. Fit logistic regression using all Lag variables + Volume
3. Confusion matrix and accuracy on training data
4. Fit using only Lag1 + Lag2 (train: 2001–2004, test: 2005)
5. Repeat with LDA, QDA, KNN ($K = 1, 3, 5$)
6. Which method gives the best test accuracy?

Key finding: QDA achieves $\approx 60\%$ accuracy

Exercise 3: Weekly Data (ISLR 4.8 Q11)

Data: `Weekly` $n = 1{,}089$ weekly returns

1. EDA: patterns in Lag1–Lag5 and Volume?
2. Fit logistic regression with all Lag variables + Volume
3. Confusion matrix — what is the overall error rate?
4. Fit using only Lag2 (train: 1990–2008, test: 2009–2010)
5. Repeat with LDA, QDA, KNN ($K = 5$)
6. Compare results — which performs best?

Exercise 4: Boston Crime (ISLR 4.8 Q16)

Data: `Boston` Create binary: `crim > median`

1. Create binary response: `high_crime <- ifelse(crim > median(crim), 1, 0)`
2. Explore predictors with boxplots and correlations
3. Fit logistic regression, LDA, KNN (try various K)
4. Which predictors are most useful?
5. Which method gives the best test accuracy?

Hint: `rad`, `tax`, `nox`, `lstat` should be strong predictors

Solution: Exercise 1 — Default

`balance`: $\hat{\beta} = 0.0057$, $p < 0.001$
`income`: $\hat{\beta} = 0.003$, $p = 0.62$
Income is not significant!

Test error $\approx 2.7\%$ (varies by split)

Adding `student` reduces error slightly but effect is small

Confusion matrix (typical):

	Pred. No	Pred. Yes
No	2,413	12
Yes	60	15

Sensitivity $\approx 20\%$
Low, because defaults are rare

Solution: Exercise 2 — Smarket

Key results (test 2005):

Method	Accuracy
Logistic (Lag1, Lag2)	56.1%
LDA (Lag1, Lag2)	56.0%
QDA (Lag1, Lag2)	59.9%
KNN $K = 1$	50.0%
KNN $K = 3$	53.6%

Insights:

- Using all Lags: no predictor is significant
- Lag1 + Lag2 only: slight improvement
- QDA best: suggests non-linear boundary
- KNN $K = 1$ = pure noise

Solution: Exercise 3 — Weekly

EDA findings:

- Volume increases over time
- Lag variables: no clear pattern
- Only Lag2 is marginally significant

Best model:

Logistic with Lag2 only: 62.5% accuracy
LDA: 62.5% (identical)

Confusion matrix (Logistic):

	Pred. Down	Pred. Up
Down	9	34
Up	5	56

Model mostly predicts "Up"
Good at catching Up days
but poor at Down days

Solution: Exercise 4 — Boston Crime

Strong predictors:

`nox`, `rad`, `tax`, `dis`, `lstat`, `age`

Best methods:

Method	Accuracy
Logistic	$\sim 87\%$
LDA	$\sim 86\%$
KNN $K = 3$	$\sim 89\%$
KNN $K = 10$	$\sim 87\%$

Insights:

- KNN with small K performs best
- Suggests non-linear boundaries
- `nox` and `rad` are the strongest single predictors
- Crime is well-predicted by socioeconomic variables

What's Next?

Week 1: Linear Reg.

↓

Week 2: Classification

↓

Week 3: Cross-Validation

Homework: ISLR Chapter 4: Q10, 11, 13, 16
Lab: `Lab_Week02.R` — Logistic, LDA, QDA, NB, KNN on Default + Smarket
Next week: How do we *properly* estimate test error? $\Rightarrow$ Cross-Validation!

Thank You

Questions?

Dr. Endri Raço — Machine Learning with R

`github.com/endri81/ISLRv2-solutions`